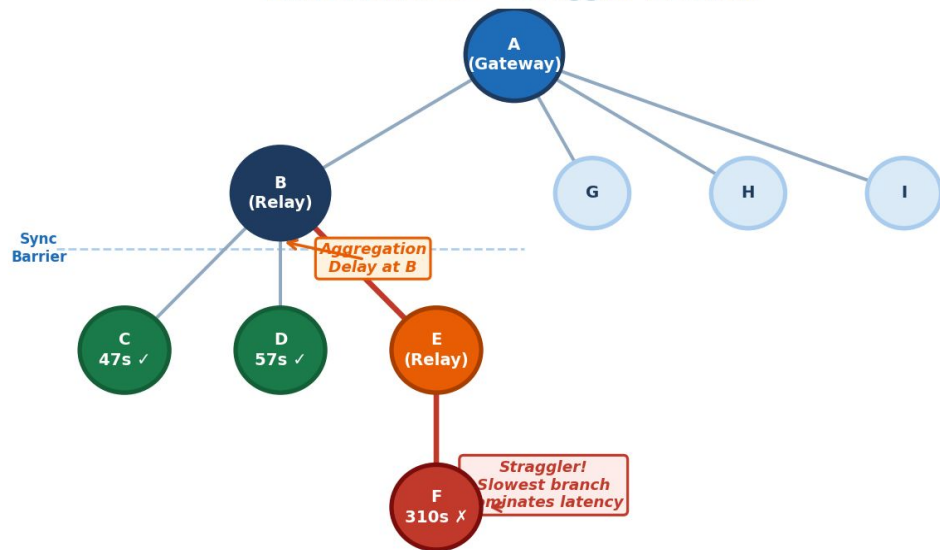
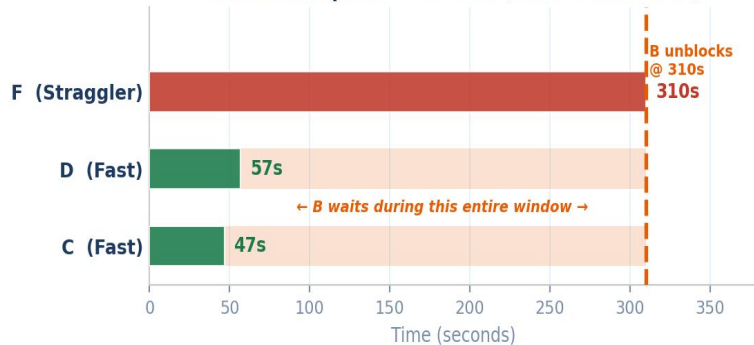


Aggregation Delay and the Straggler Problem in Distributed Query Execution

Execution Tree — Straggler Problem



Worker Completion Timeline (Node B's children)



Why Aggregation Delay Occurs:

- Relay nodes must collect all children before returning upstream
- Partial results cannot propagate — each level is a sync barrier
- Slowest node (straggler) determines total branch latency

310s dominates over **47s** & **57s** — fast nodes idle while B waits

Single-Node: Sequential scan, no sync cost

VS

Distributed Tree: Parallel execution + synchronization delay — slowest branch adds waiting cost

Distributed query latency is bounded by the slowest worker due to aggregation delay and synchronization barriers.